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# MIS 522 - Practicum Assignment

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## Group Questions - 1

1. Why is Pandas useful for MIS data analysis?

Pandas is useful for MIS data analysis because it helps us efficiently organize, clean, filter, manipulate and analyze large datasets using tables (Data Frames).

2. Why do we use NumPy?

NumPy is used for fast numerical calculations and array operations. It provides efficient mathematical functions that are useful for data analysis and statistics.

3. Why are visualizations important for managers?

Visualizations help managers understand complex data quickly, identify trends and patterns, compare performance and make better business decisions.

4. Why is reproducibility important when creating a teaching dataset?

Reproducibility ensures that a teaching dataset and its results can be created and verified again using the same steps and data-generation process. This makes the analysis reliable, consistent and easier for students to learn from.

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## Group Questions - 2: Table 1

1. Why are there more than 3000 rows?
Table 1 contains 25 duplicate household records. They were intentionally added to the original 3000 records, making the raw table 3025 rows.

2. Which variables contain NULL values?
The variables containing NULL values are:
Employment_status : 55
Education : 35
Monthly_Income : 78

3. Which values are logically invalid?
Negative values in Monthly_Income and Household_size are logically invalid.

4. Why is negative income a data-quality problem?
It is logically invalid and may distort financial analysis.

5. Why can inconsistent province spelling affect reports and dashboards?
Inconsistent province spelling can affect reports and dashboards as it can create duplicate categories and inaccurate summaries.

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# Group Questions -3 : Table 2

1. Why should expenditure normally not be negative?

Expenditure represents money spent by households so it is normally recorded as a non-negative value. A negative value may indicate a data-entry or coding error.

2. What is an orphan Household-ID?

An Orphan Household-ID is an ID that appears in the dataset but doesnot have a corresponding household record in the cleaned household table.

3. How can duplicate monthly observations affect the analysis?

Duplicate observations can give excessive weight to certain households or months, thereby affecting totals, averages, and overall trends.

4. Which variables can be considered time-series variables

Variables such as Observation-Month, Inflation-Rate, Food-Inflation, Interest-Rate, Essential-Expenditure, Discretionary-Expenditure

and Employment_Index can be analyzed and considered time-series variables because their values can be observed and compared across different months. They help identify changes, trends and patterns over time.

5. Why should extreme expenditure values be investigated?

Extreme values may represent genuine high expenditure or data errors. Investigating them helps ensure that unusual observations do not incorrectly influence the analysis.

## Group Questions -4

1. Audit Table

| Problem | Variable | Count | Treatment |
|---|---|---|---|
| Missing values | Employment_Status, Educatin | 55<br>35 | Fill with most common val |
| | and Monthly_Income | 89 | Fill with median |
| Duplicate rows | Household_ID (whole row) | 25 | Remove duplicates |
| Negative Income | Monthly_Income | 20 | Treat as invalid and fix |

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# Audit Table

| Problem | Variables | Count | Treatment |
|---|---|---|---|
| Negative household size | Household-size | 12 | Treat as invalid and fix. |
| | | | |
| | | | |

Q.1. Which problem should be fixed first?

Negative values (Negative income and Negative household size) should be fixed first because they are invalid data and can affect the analysis.

Q.2. Which cleaning decisions require business knowledge?

Decisions about missing income, employment status, education and household size require business knowledge.

Q.3. Why should the dirty raw data be preserved?

It should be preserved for backup, checking and tracking what was changed during cleaning.

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## Group Questions - 5

1. Which columns have missing values?
   Inflation_Rate (121), Essential_Expenditure (102), and Discretionary_Expenditure (70) have missing values

2. Which contain negative values?
   Inflation_Rate (7), Interest_Rate (8), Essential_Expenditure (10) and Discretionary_Expenditure (8) contain negative values.

3. Which negative values are invalid for this case?
   Negative inflation, interest rate and expenditures are invalid because these values should not normally be negative in this dataset.

4. Which variables should be checked for outliers?
   Inflation_rate, Interest_rate, Essential_expenditure and Discretionary_Expenditure should be checked because they contain unusual values.

5. What could happen if this dirty data were loaded directly into Power BI?
   It could produce incorrect charts, calculations and misleading business insights.

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# Group Questions - 6

1. Is the cleaned master table exactly 8000 households?

Yes, the cleaned master table has exactly 8000 households.

2. Why was negative income converted to missing before imputation?

Negative income is invalid, so it was treated as missing before filling the value.

3. Why use median for income?

Median is less affected by extreme income values, so it gives a more reliable replacement

4. Why use mode for categorical variables?

Mode is the most common category so it is suitable for filling missing categorical values.

5. Why must Household_ID be unique in the

Each Household_ID should represent one household. Duplicates could cause incorrect analysis and double counting.

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# Group Questions-7

1. How many orphan observations were removed?

80 orphan observations were removed because their Household-ID didnot match any household in master table.

2. Why should orphan records be removed before merging?

Orphan records have no matching household information. Keeping them can create incomplete or incorrect records after merging.

3. Why flag outliers instead of automatically deleting them?

Some outliers may be genuine and important observations. Therefore, they should be flagged and investigated before deciding whether to remove them or not.

4. What is the danger of replacing extreme values with the mean?

Replacing extreme values with the mean can hide important information and can reduce the actual variation in the data.

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## Group Questions - 8

1. Is income skewed?

Yes, Income is positively skewed because a few households have very high income values.

2. Is expenditure skewed?

Yes, expenditure is also skewed because some households have unusually high expenditure.

3. Which variable has stronger evidence of extreme observations?

Income show stronger evidence of extreme observations because it has more widely spread values and very high maximum values.

4. Why is median useful for skewed economic data?

Median is useful because it is less affected by extreme values and gives a more reliable picture of the typical household.

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# Group Questions - 9

1. Which province has the largest sample?

Bagmati has the largest sample with 726 households.

2. Which employment category is largest?

Employed is the largest employment category i.e. 1313.

3. Is the dataset balanced across provinces?

No, the dataset is not balanced because the number of households differs across provinces.

4. Why can unequal group size affect interpretation?

Unequal group sizes can make some provinces or groups more influential, which may lead to biased or misleading comparisons.

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Group Questions - 10

1. Compare mean and median income.

Mean income is higher than median income. Mean income is Rs 74,077.07 while median income is Rs 66,333.00. It is higher by about 11.7%.

2. What does the difference tell us?

It shows that income is positively skewed, with some higher-income households pulling the mean upward.

3. Which variable has the greatest standard deviation?

Monthly Income has the greatest standard deviation of 36,850.33. This indicates greater variation in income.

4. Which variables are suitable for mode analysis?

Province and Employment Status are suitable because they are categorical variables.

5. Why should managers not rely only on average values?

Average values can be affected by

extreme values. So, managers should also consider the median and variation for better decisions.

Group Questions - 11

1. Which province has the highest average income?

Sudurpashchim has the highest average income of Rs 76,462.22. It has the highest average among all seven provinces.

2. Which has the highest median income?

Bagmati has the highest median income (66,442). Most other provinces have a median of about (66,333), so the differences are small. Gandaki is lowest at 66129 and the remaining provinces share 66,333.

3. Are the ranking identical?

No, Sudurpashchim ranks first on average income, but Bagmati ranks first on median income. The difference arises because the average is pulled up by a few high earners, while the median reflects the typical household. Sudurpashchim's high average therefore doesnot mean most of its households earn more.

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4. Which province has the largest average household size?

Lumbini has the largest average household size at 4.59 members, followed by Bagmati (4.55) and Koshi (4.51). Gandaki has the smallest at 4.35. Sizes falls in a narrow range, so family-size products would suit Lumbini and Bagmati best.

5. What managent decision could use this summary?

A consumer goods company could use this summary to choose launch market and pricing. Sudurpashchim and Madhesh (highest average income) suit premium products, while Lumbini and Bagmati (largest households) suit family-size packs. Bagmati, with the highest median income, is a strong first market.

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# Group Questions - 12

1. Why is the merged table larger than 8,000?

Table 2 has repeated monthly observations for each household, and the merge keeps every Table 2 row. The merged table therefore has 7970 rows, much larger than the roughly 8000 unique household in Table 1.

2. Explain the one-to-many relationship

One household in Table 1 (One Household_ID) can appear in many rows of Table 2, one per month. Each Table 2 row belongs to only one household. So, the household details (province, areatype, income) repeat across that household's monthly rows.

3. Why is many-to-one appropriate?

Table 2 (left) has many rows per Household_ID, while Table 1 (right) has one row per Household_ID after duplicates were removed. Many-to-one relationship states this expected structure and pandas raises an error if the right table has duplicate keys.

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4. What could happen if Household-ID were duplicated in Table 1?

Each Table 2 row would match more than one Table 1 row, so rows would multiply and the merged table would inflate. This would double-count households and distort averages and totals. Conflicting household details could also be attached to the same record.

5. Why is merge validation important in MIS?

Validation catches key problems, such as duplicates and wrong relationships, before they corrupt the data. In this merge, a left join left missing values in Employment_Status (133), Education (81) and Monthly_Income (218), which shows why checking after a merge is essential. Reliable merged data gives managers trustworthy reports and decisions.

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Group Questions - 13

Variables

1. Savings_Estimate
- Formula: Monthly_Income - Total_Expenditure
- Measures: The money a household has left after spending, or a deficit if negative.
- Added value: Income alone doesnot show wheather a household can save or not. HH101667 earns Rs 35,816 but spends Rs 43083, giving (-7,267) savings.
- Management use: A bank can target savings products at households with high savings.

2. Savings_Rate
- Formula: Savings_Estimate / Monthly_Income
- Measures: The share of income a household saves.
- Added value: It lets households of different income be compared fairly. HH102657 saves about 76% of income while HH101667 has -20%.
- Management use: Financial institutions can design deposit schemes for strong savers.

3. Financial_Burden
- formula: Total Expenditure / Monthly_Income
- Measures: The share of income consumed by spending. A value above 1 means spending exceeds

income.

- Added value: It shows financial pressure that income alone hides. HH101667 has a burden of 1.20 (High), while HH102657 has 0.24 (low)
- Management use: Lenders can assess repayment risk before approving loans.

4. Income_Per_Person
- Formula: Monthly_Income / Household_Size
- Measures: Income available to each household member.
- Added value: Total income ignores dependents. HH101985 earns ₹74,674 but has only Rs 12,445.67 per person, while HH101850 earns a similar Rs 77,774 and has Rs 38,887 per person.
- Management use: Companies can set pricing and pack sizes by purchasing power per person.

5. Real_Income
- Formula: Monthly Income ÷ (1 + Inflation Rate/100)
- Measures: Income adjusted for inflation, which is its actual purchasing power.
- Added value: Nominal income can hide falling purchasing power. HH101850's 77,774 is worth about 72,966 in real terms

- Management use: Employers and policymakers can review wage adjustments and cost-of-living support.

Key discussions

Income alone doesnot show how much a household spends, how many people it supports, or how much prices have risen. For eg, HH101850 and HH101925 earn almost the same (Rs 77,774 and Rs 74,674), but their income per person is Rs 38,887 and Rs 12,445.67 because household sizes differ. Their spending and saving also differ, so one household can save comfortably while the other is under pressure.

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Group Questions - 1:4

1. Which income group spends the most?
Low Income spends the most at Rs37,378.86 on average, but the three groups are almost identical (Middle: Rs 37,862.41, High: Rs 37,815.67). Spending barely changes with income.

2. Which has the highest financial burden?
Low Income has the highest burden at 0.99. Middle Income is 0.57 and High Income is 0.35. Low-Income households spend almost all of their income.

3. Does higher income automatically mean lower burden?
Yes, burden falls as income rises (0.99, 0.57, 0.35) because spending stays almost flat while income rises. This is not automatic for every household, though. A high earner with large expenses or many dependents can still carry a high burden.

4. What might explain a high burden?
A high burden comes from low income relative to essential cost such as

food and housing. large household size, inflation and rising interest rates on loan can also raise it.

5. Which group would you monitor as a bank or financial manager?

Low Income because their average savings are only ₹2984.33 and their burden is 0.99, so they have little buffer and the highest rick of miseing loan repayments.

Group Questions -15

1. What trend do you observe?

All three indicators stay fairly stable from 2024-01 to late 2025, with no strong rise or fall. Inflation stays around 6.2%, food inflation around 7.2 to 7.5%, and the interest rate around 7.5%, with a slight dip towards the end.

Does food inflation behave differently from overall inflation?

Yes, food inflation is consistently higher (about 7.4% against 6.2%) and moves up and down more from month to month. Food prices are rising faster than general prices.

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3. Why should monitor inflation and interest rates together?

Together they show the real cost of borrowing. Here the interest rate is only about 1.8 points above inflation, so the real return and real loan cost are small. Managers need both to judge pricing, loans and repayment capacity.

4. What happens if expenditure rises faster than income?

If expenditure rises faster than income, saving falls and financial burden rises. If burden goes above 1, households run deficits, borrow more and risk defaulting.

5. What additional economic variable would improve the analysis?

Income or wage growth rate would improve analysis, since it can be compared directly with inflation to show real purchasing power. The unemployment rate would also help.

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## Group Questions - 16.

1. Identify two strong relationships.

- Monthly Income and Saving Estimate (0.92): Household with higher income save much more!

- Savings Estimate and Financial Burden (-0.78): Higher savings go with lower burden.

2. Identify one weak relationship.

The one weak relationship is Monthly Income and Total Expenditure (0.00). Spending does not rise with income, which matches the earlier finding that all income groups spend about the same. Inflation rate and Interest rate (about 0.00) is another weak pair.

3. Monthly Income and Financial Burden (-0.60) are the most useful relationship. It shows that lower-income households carry the heaviest burden, so banks and businesses can target credit limits, risk monitoring and pricing by income group.

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4. Correlation shows that two variable move together, not that one causes the other. A third factor may drive both, or the link may be coincidence. Savings and burden are also correlated partly by construction, because both are calculated from income and expenditure.

5. Other variables that could explain the relationships are Education, employment status, household size area type, debt and loan obligations and wage growth.

Group Questions - 17.

1. High Income group shows greater expenditure variation, with the most outliers, though all three groups are similar. Most of the variation comes from extreme values.

2. A mean gives only one central value. The boxplot also shows the median, the spread of the middle 50% of households and the outliers. Here it shows that most households spend within a narrow range while a few spend far more, which the mean would hide.

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3. NO, Outliers may be genuine observations, and deleting them loses useful information. Managers should first check whether each is a data error or a real value. Only errors should be removed or corrected, and genuine outliers should be kept or analysed separately.

4. A household may have a one-off large expense such as medical treatment, tuition fees or a weddings. A large household with many dependents can also have much higher essential spending.

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## Group Questions 18

1. Bagmati has highest median household income with median of about 66000 to 67000. Gandaki is lowest at about 58000, and the others fall in between (Sudurpashchim, Lumbini, Madhesh, Koshi, Karnali).

2. Madhesh and Bagmati provinces shows the greatest income variation. Madhesh has the tallest box (about 46000), and Bagmati is close Behind. Bagmati also has the largest overall spread because of its highest outlier (about 820000)

3. Yes, Bagmati has the most extreme income values at about 820000 and Gandaki, Sudurpashchim and Koshi go beyond 250000.

4. Boxplot is useful because it shows the median, spread, whiskers and outliers for all provinces side by side, so they can be compared quickly. It also reveals skewness and extreme values that a single average hides.

5. A few very high earners pull the average above what a typical household earns. Here

a province can show a high average while most households earn much less, so the median and spread should be read alongside the average.

6. Industry and job mix, urban vs rural location, education levels, employment opportunities, remittances and infrastructure could explain the differences. Cost of living and local prices change real income as well.

Group Questions - 19

1. Rural areas have a higher average income (Rs 74,347.26) than urban areas (Rs 73,281.33)

2. Rural areas have a slightly higher average expenditure (Rs 37,545.29) than urban areas (Rs 37,217.28)

3. Rural areas have higher average savings (Rs 86,801.97) compared to urban areas (Rs 86,064.55)

4. Urban areas have a slightly greater financial burden, with an average burden of 0.64 compared to 0.68 in rural areas.

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5. Differences may be explained by variations in living costs, employment opportunities, wages, access to resources and household spending patterns. Urban areas may have higher living costs while rural households may have different come and expenditure structures.

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